Technical University of Denmark

46755 Renewables in electricity markets,

Assignment 1: Market Clearing (System Perspective)

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Group 8

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Table 1: Contribution of each member to project tasks. Repository: GitHub link

	Emirhan	Heba	Teodor	Quoc
Contribution to model development	25%	25%	25%	25%
Contribution to programming	25%	25%	25%	25%
Contribution to the analysis of numerical results	25%	25%	25%	25%
Contribution to writing the report	25%	25%	25%	25%

Contents

Nomenclature

Sets and Indices

- $g \in G$: Set of conventional generators.
- $w \in W$: Set of wind power plants.
- $d \in D$: Set of demands.
- $t \in T$: Set of time periods (hours $1 \ldots 24$).
- $i, j, k \in N$: Set of nodes (buses) in the network.
- $A, B \in Zone$: Set of zones in the network.
- $g, w, d \in \Psi$: Set of generations, wind farms, and load in a Zone.
- $g \in G_{ser}$: Set of generators participating in the balancing and reserve markets.

Parameters

- $\overline{P}_g^G$: Maximum power of conventional generation g [MW].
- $\overline{P}_{w,t}^W$: Forecasted power available from wind farm w at time t [MW].
- $\overline{P}_{d,t}^D$: Maximum power consumption of demand d at time t [MW].
- C_g: Marginal cost of conventional generator g [€/MWh].
- U_d: Bid price of demand d [€/MWh].
- η_{ch}, η_{dis}: Charging and discharging coefficient of battery energy storage system (BESS).
- $\underline{SOC}^{BESS}, \overline{SOC}^{BESS}$: Minimum and maximum State-of-charge (SOC) of BESS [%].
- $\overline{P}_{i,j}$: Capacity of transmission line ij [MW].
- $ACT_{A,B}$: The Available Transfer Capacity between Zone A and B [MW].
- ΔP: Power imbalance in the balancing market [MW].
- P_g^{DA}: DA power of generation g[MW].
- C_g^u, C_g^d: Upward and downward reserve capacity cost of generator g [€/MWh].
- R_g^+, R_g^-: Maximum up and down reserve capacity of generator g [MW].

Variables

- $p^G_{g,t}$: Power produced by conventional unit g at time t [MW].
- $p^W_{w,t}$: Power produced by wind farm w at time t [MW].
- $p^D_{d,t}$: Power consumed by load d at time t [MW].
- $p^{BESS}_{ch,t}, p^{BESS}_{dis,t}$: Charging and discharging power of BESS at time t [MW].
- SOC^{BESS}_t: SOC of BESS at time t [%].
- $\theta_{n,t}$: Voltage angle at bus n at time t [rad].
- $f_{A,B,t}$: Power exported from Zone A to Zone B at time t [MW].
- $p^{G\uparrow}_g, p^{G\downarrow}_g$: Upward and downward regulation power of generator g at time t [MW].
- $p^{D,\text{curt}}_d$: Curtailed power of demand d [MW].
- $r^u_{g,t}, r^d_{g,t}$: Upward and downward reserve power of generator g [MW].

0 Introduction

This report addresses the fundamental challenges of clearing electricity markets under high renewable energy penetration and technical network constraints. The objective is to analyze a series of optimization models across six steps: (**1-2**) uniform pricing in single and multi-hour settings with energy storage; (**3**) nodal and zonal pricing under network congestion; (**4**) derivation of KKT conditions to bridge centralized optimization and decentralized equilibrium; and (**5-6**) balancing and reserve market settlement schemes.

0.1 Case Study and General Assumptions

The analysis is implemented in an IEEE 24-bus transmission system [1]. The following assumptions maintain mathematical consistency throughout the study. First, regarding **Bidding Behavior**, all producers are price-takers bidding marginal costs (zero for renewables). Second, **Demand** is modeled as price-elastic with bid prices set high enough to ensure maximum fulfillment. Lastly, technical parameters for storage and the network are consistent with the values established in Section 2.

1 Copper-plate Single Hour

The single-hour market-clearing problem is analyzed under the copper-plate assumption. This means that the transmission network is neglected, and the entire power system is treated as one unconstrained market area. As a result, electricity can be exchanged freely between all generators and loads, and a single uniform market-clearing price (MCP) is established for the whole system. The market is modeled as a social welfare (SW) maximization problem. Conventional generators submit offers equal to their marginal production costs, while renewable generators are assumed to bid at zero marginal cost. On the demand side, consumers are represented by price-elastic demands with given bid prices and maximum consumption levels. The model determines the intersection of supply and demand curves to find the MCP, SW, and the profit of each market participant.

1.1 Mathematical Model

The market-clearing for a single-hour copper-plate system ($t = 1$) is formulated as a Social Welfare (SW) maximization problem:

$$\max \sum_{d} U_d p^D_{d,t} - \sum_{g} C_g p^G_{g,t} - \sum_{w} 0 \cdot p^W_{w,t} \tag{1}$$

Subject to (for all g, w, d):

$$0 \leq p^G_{g,t} \leq \overline{P}^G_g \qquad (\underline{\mu}^G_{g,t}, \overline{\mu}^G_{g,t}) \tag{2}$$

$$0 \leq p^W_{w,t} \leq \overline{P}^W_{w,t} \qquad (\underline{\mu}^W_{w,t}, \overline{\mu}^W_{w,t}) \tag{3}$$

$$0 \leq p^D_{d,t} \leq \overline{P}^D_{d,t} \qquad (\underline{\mu}^D_{d,t}, \overline{\mu}^D_{d,t}) \tag{4}$$

$$\sum_{g} p^G_{g,t} + \sum_{w} p^W_{w,t} - \sum_{d} p^D_{d,t} = 0 \qquad (\lambda_t) \tag{5}$$

The objective function (1) maximizes the total surplus, while constraints (2) – (4) enforce the physical limits of generators, wind farms, and price-elastic demands. Equation (5) ensures the power balance, where the associated dual variable λ_t represents the MCP.

1.2 Market Clearing Results

The optimization problem was solved to optimality and produced the following results for hour $t = 1$:

The objective value reported by the solver corresponds directly to the total SW of the system, confirming that the optimization problem has been correctly formulated as a welfare maximization problem. The MCP of 6.02 €/MWh represents the uniform price paid by consumers and received by producers in the copper-plate market. Economically, this price is determined by the marginal accepted generator, meaning the last generator required to satisfy total system demand. Generators with marginal costs below the MCP are dispatched, while generators with higher marginal costs are not selected in the optimal solution. The total operating cost of €2,237.42 represents the sum of production costs of all dispatched generators. The total SW obtained in the optimization is €83,122.39. SW represents the overall economic benefit created in the market and is defined as the difference between the total willingness to pay of consumers and the total production cost of generators. A high SW value indicates that a large amount of demand is served at relatively low production costs compared to consumers' bid prices.

The profit of each producer and the utility of each demand are calculated as

$$\Pi_g = p_g(\lambda - C_g) \tag{6}$$

$$Utility_d = p_d(U_d - \lambda) \tag{7}$$

Table 2 shows the profit of each producer, including both wind farms and conventional generators. As seen, the profit for the wind farms is considerably higher than that of the conventional units. This is primarily attributed to their low operating costs, as renewable units are assumed to have a marginal cost of zero. In contrast, most conventional units do not generate a profit because they don't enter the market, with the notable exceptions of generators 9 and 10. Consumers obtain positive utility as shown in Table 3 whenever their bid price exceeds the MCP.

Table 2: Total Profit of Suppliers (Wind farms and conventional units)

Supplier	Total Profit of Suppliers [€]
W1	1,027.87
W2	1,109.35
W3	1,121.06
W4	1,074.31
W5	1,012.83
W6	1,081.69
G1	0.00
G2	0.00
G3	0.00
G4	0.00
G5	0.00
G6	0.00
G7	0.00
G8	0.00
G9	220.00
G10	1,806.00
G11	0.00
G12	0.00

Table 3: Utility of all demand

Demand	Utility of demand [€]
B1	6,216.69
B2	5,205.76
B3	9,004.04
B4	3,434.23
B5	3,042.83
B6	5,353.97
B7	4,496.45
B8	5,581.09
B9	5,133.72
B10	5,094.68
B13	6,073.17
B14	3,835.57
B15	5,289.66
B16	1,363.34
B18	3,554.34
B19	1,407.51
B20	582.22

1.3 The KKT conditions

The MCP can also be verified using the Karush-Kuhn-Tucker (KKT) conditions of the optimization problem. In particular, the dual variable associated with the system power balance constraint represents the shadow price of electricity, which corresponds to the MCP. The following are their optimization problems:

$$\max_{p_g^G} p_g^G(\lambda - C_g) \qquad \max_{p_d^D} \quad p_d^D(U_d - \lambda) \tag{8}$$

$$\text{subject to:} \quad 0 \leq p_g^G \leq P_g^G \ : \ \underline{\mu}_g^G, \ \overline{\mu}_g^G \qquad \text{subject to:} \quad 0 \leq p_d^D \leq P_d^D \ : \ \underline{\mu}_d^D, \ \overline{\mu}_d^D \tag{9}$$

The problems are bound together with the following balance equation:

$$\sum_{d \in D} p_d^D - \sum_{g \in G} p_g^G = 0 \qquad : \lambda \tag{10}$$

The Lagrangian function is derived:

$$\begin{aligned} \mathcal{L}(p_g^G, p_d^D, \lambda, \mu_{g,d}) = & \sum_{g \in G} \left(p_g^G(\lambda - C_g) - \overline{\mu}_g \, p_g^G + \underline{\mu}_g (p_g^G - P_g^G) \right) \\ & + \sum_{d \in D} \left(p_d^D(U_d - \lambda) - \overline{\mu}_d \, p_d^D + \underline{\mu}_d (p_d^D - P_d^D) \right) + \lambda \left(\sum_{d \in D} p_d^D - \sum_{g \in G} p_g^G \right) \end{aligned} \tag{11}$$

The optimality Karush–Kuhn–Tucker (KKT) conditions for one generator are:

$\nabla_{p_g^G, p_d^D} \mathcal{L} = 0$	(stationarity)	$h(p_g^G, p_d^D) = 0$	(primal feasibility)
$0 \leq -g(p_g^G, p_d^D) \perp \mu \geq 0$	(complementarity)	λ free	(dual feasibility)

To verify the KKT conditions for hour $t = 1$, the optimal dispatch is examined. The total served demand of 1775.835 MW is mentioned in Figure B.2. The dispatched power of conventional units and wind units is:

$$P_{g8} = 8.21 \text{ MW}, \qquad P_{g9} = 400 \text{ MW}, \qquad P_{g10} = 300 \text{ MW}.$$

$$P_{w1} = 170.74 \text{ MW}, \ P_{w2} = 184.28 \text{ MW}, \ P_{w3} = 186.22 \text{ MW},$$

$$P_{w4} = 178.46 \text{ MW}, \ P_{w5} = 168.24 \text{ MW}, \ P_{w6} = 179.68 \text{ MW}.$$

The total generation is 1,775.83 MW, which matches the total served demand of 1,775.83 MW, thus satisfying the balance constraint. Since generator $g8$ is dispatched strictly within its capacity limits, it is the marginal generator. For generator $g8$ both complementary slackness conditions yield:

$$0 \leq (\overline{P}_8^G - p_8^G) \perp \overline{\mu}_8^G \geq 0 \Rightarrow \overline{\mu}_8^G = 0$$

$$0 \leq (p_8^G) \perp \underline{\mu_8^G} \geq 0 \Rightarrow \underline{\mu_8^G} = 0$$

Inserting these values into the stationarity condition of the Lagrangian gives:

$$\frac{\partial \mathcal{L}}{\partial P_8^G} = C_8^G + \overline{\mu}_8^G - \underline{\mu_8^G} - \lambda = 0 \Rightarrow C_8^G = \lambda \Rightarrow \lambda = 6.02 \text{ €/MWh}$$

2 Copper-Plate, Multiple Hours

In this step, the copper plate model from Step 1 is extended to consider multiple time periods (24 hours). A large-scale battery energy storage system (BESS) is also taken into account to enforce intertemporal constraints. The data for this BESS are referred to from [2]. Moreover, to make the problem simple, it is assumed that ESS submits zero bids and offers prices to the market.

2.1 Mathematical Model

The constraints in this step are updated as follows:

$$\sum_g p_{g,t}^G + \sum_w p_{w,t}^W + (p_{dis,t}^{BESS} - p_{ch,t}^{BESS}) - \sum_d p_{d,t}^D = 0 \quad \forall t \tag{12}$$

$$\underline{SOC}^{BESS} \leq SOC_t^{BESS} \leq \overline{SOC}^{BESS} \quad \forall t \tag{13}$$

$$SOC_t^{BESS} = SOC_{t-1}^{BESS} + (p_{ch,t}^{BESS} \cdot \eta_{ch} - \frac{1}{\eta_{dis}} p_{dis,t}^{BESS}) \frac{1}{E^{BESS}} \quad \forall t \tag{14}$$

$$0 \leq p_{ch,t}^{BESS} \leq \overline{P}_{ch}^{BESS} \quad \forall t \tag{15}$$

$$0 \leq p_{dis,t}^{BESS} \leq \overline{P}_{dis}^{BESS} \quad \forall t \tag{16}$$

$$SOC_{24}^{BESS} = SOC_0^{BESS} \tag{17}$$

At this time, the power balance demonstrated in Constraint (12) has the BESS component. Constraint (13) shows the upper and lower limits of the State-of-charge (SOC). In this work, the upper and lower bounds of BESS are set at 100% and 0%, respectively [2]. The SOC of BESS can be determined through Equation (14), with η_{ch} and η_{dis} being the charging and discharging efficiencies. To avoid simultaneous charging and discharging, η_{ch} and η_{dis} are set at 0.83 and 0.85 [2], respectively. These values are standard benchmarks for AC-to-AC conversion losses and auxiliary power requirements in grid-scale storage [3]. Equations (15) and (16) ensure ESS operating in the allowable range. To ensure a sustainable daily operation, a circularity constraint (17) is enforced. While various sizes were evaluated, the results below focus on the **3,000 MW / 6,000 MWh** configuration, which represents the optimal capacity for this system, as explained why it is optimal below.

2.2 Results and Discussion

2.2.1 System Economic Impact

The introduction of the optimal BESS yielded a total system operating cost of €221,261.69 and a total SW of €2,310,519.50. By shifting energy from low-cost off-peak hours to high-cost peak hours, the BESS effectively reduces overall system costs and maximizes economic benefit.

2.2.2 Market Clearing Price Dynamics

Figure 1 illustrates the impact of the BESS on the MCP. It should be noted that different BESS sizes (1,000MW/2,000MWh, 3,000MW/6,000MWh, and 30,000MW/ 60,000MWh) are used for sensitivity analysis. In the baseline scenario without storage, prices exhibit high volatility, driven by the marginal costs of the specific generators required to satisfy peak demand.

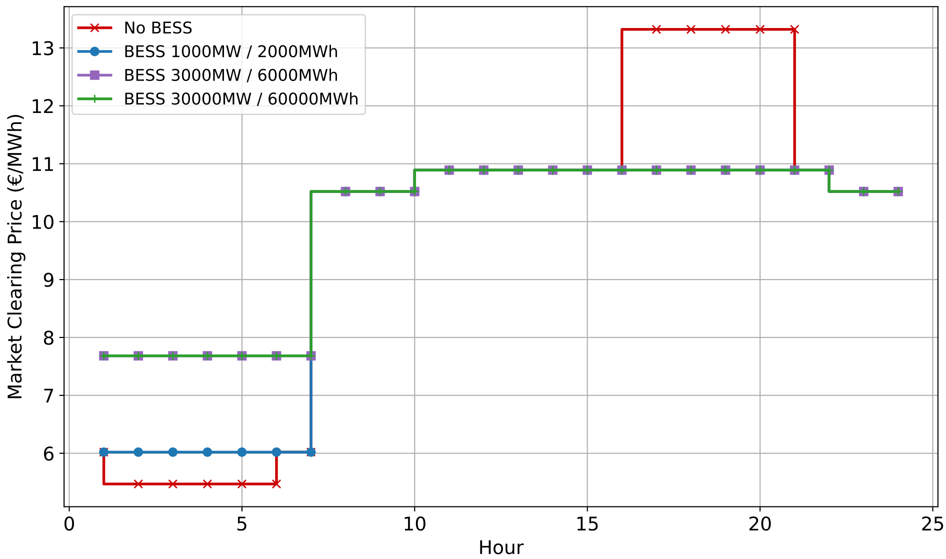

Figure 1: Hourly Market Clearing Prices: Comparison between baseline and BESS scenarios

With the implementation of the optimal BESS, the price curve flattens significantly. The battery charges during off-peak hours and discharges to supply necessary energy throughout the peak period. During these hours, the MCP is determined not by a conventional generator's marginal cost, but by the opportunity cost of the stored energy.

2.2.3 Operational Profile: State of Charge and Generation

The physical mechanism of this energy arbitrage is visualized in Figures 2 and 3. The unit charges during the morning (hours 1–7), coinciding with high wind production and low demand, and discharges during the evening peak (hours 17–22). The 3,000 MW system fully utilizes its capacity to cover the entire peak duration. In contrast, the smaller 1,000 MW system reaches its energy limit prematurely, leaving the network exposed to subsequent price spikes, leaving the system unable to suppress subsequent price spikes.

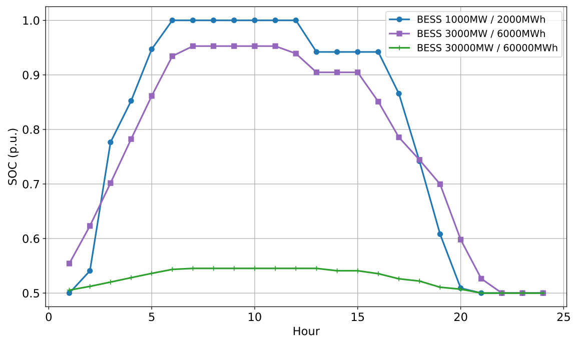

Figure 2: State of Charge (SOC) profiles

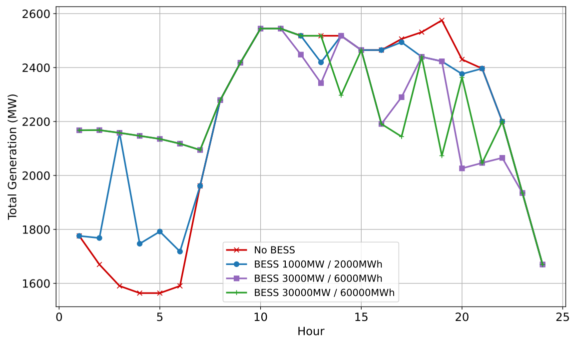

Figure 3: Total system generation profiles

2.2.4 Market Participant Profitability

Table 4 summarizes the profits of the market participants. Notably, the BESS earns exactly €0.00 in profit. This is a standard outcome in centralized SW maximization where BESS submits zero bids, the storage is dispatched to maximize SW rather than its own profit, resulting in zero economic rent for the storage operator.

Table 4: Total 24-hour Profit for Market Participants (3,000 MW / 6,000 MWh)

Unit	Profit [€]	Unit	Profit [€]	Unit	Profit [€]
G1-G5, G12	0.00	G10	71,118.08	W3	29,707.89
G6-G7	688.20	G11	1,376.40	W4	23,543.61
G8	37,032.11	W1	24,141.60	W5	19,891.76
G9	42,312.11	W2	34,359.28	**BESS**	**0.00**

2.2.5 Sensitivity Analysis: Defining Optimality

A sensitivity analysis was conducted to determine the optimal ESS capacity, as shown in Figure 4. While the SW increases sharply up to 3,000 MW, it plateaus immediately thereafter. Expanding the battery to a 30,000 MW scenario yields identical welfare, demonstrating that oversizing the system provides no additional economic benefit. Furthermore, a project of such scale would require significant capital investments and is practically infeasible.

2.2.6 Computational Aspects

The 24-hour copper-plate model was formulated as a Linear Programming (LP) problem, comprising 912 continuous variables and 960 linear constraints. Solved via Gurobi, the optimal solution was achieved in under 0.1 seconds, confirming the computational efficiency of the formulation.

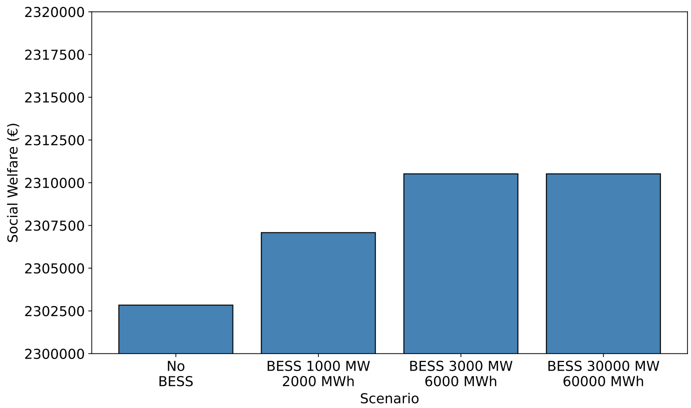

Figure 4: Sensitivity of total social welfare to BESS capacity

3 Network Constraints

The network constraints are considered in this Step, and the problem is formulated as in Step 1.

3.1 Nodal System

In this model, each node is connected to other nodes via transmission lines. In this case, instead of transmitting without any limit (Copper-plate model), several line constraints have to be considered to ensure the practical aspects of the power system. To simplify the problem, the DC approximation method is applied, but the accuracy of the obtained solutions is still ensured. Therefore, the power transmitted in a line can be represented as $p_{i,j} = B_{i,j}(\theta_i - \theta_j)$. The objective function (1) is now considered the voltage angle of each node as a decision variable:

$$\max_{p^G_{g,t}, p^D_{d,t}, \theta_{n,t}} \sum_{d,t} U_d \cdot p^D_{d,t} - \sum_{g,t} C_g \cdot p^G_{g,t} \tag{18}$$

The power balance equation 5 now has to consider the power injected from bus j to bus i $p_{j,i}$ and the power leaving bus i to bus k $p_{i,k}$. Moreover, this must be implemented on each bus i.

$$\sum_g p^G_{g,t} + \sum_w p^W_{w,t} + \sum_{j:j\to i} p_{j,i,t} - \sum_{k:i\to k} p_{i,k,t} - \sum_d p^D_{d,t} = 0 : \quad \lambda_t \quad \forall i, t = 1 \tag{19}$$

To ensure the safety and security of the power grid, the power flowing and the voltage angles must comply with the following constraints:

$$-\overline{P}_{i,j} \le p_{i,j,t} \le \overline{P}_{i,j} \quad \forall i, t = 1 \tag{20}$$

$$-\pi \le \theta_i, t \le \pi \quad \forall i, t = 1 \tag{21}$$

Moreover, it is necessary to set a reference phase angle of 0. Hence, the voltage angle at bus 13 is selected as the reference $\theta_{13,t} = 0$ [1].

3.2 Zonal Model

In this model, individual power-line flows are neglected, and the network is partitioned into several zones, each with a set of buses. Therefore, the objective function of the zonal market prices is as follows.

$$\max_{p^G_{g,t},\, p^D_{d,t},\, f_{A,B,t}} \sum_{d,t} U_d \cdot p^D_{d,t} - \sum_{g,t} C_g \cdot p^G_{g,t} \tag{22}$$

$$\sum_{g\in\Psi_A} p^G_{g,t} + \sum_{w\in\Psi_A} p^W_{w,t} - \sum_{B:A\to B} f_{A,B,t} - \sum_{d\in\Psi_A} p^D_{d,t} = 0 : \quad \lambda_t \quad \forall A, t = 1 \tag{23}$$

$$-ACT_{B,A} \leq f_{A,B,t} \leq ACT_{A,B} \quad \forall A, \forall B, t = 1 \tag{24}$$

$$f_{A,B,t} = -f_{B,A,t} \quad \forall A, \forall B, t = 1 \tag{25}$$

The power balance constraint in Zone A is described in Constraint (23) with Ψ_A is the set of generations, wind farms, and loads in Zone A. Constraint (24) is used to restrict the power transferred between two regions, and Constraint (25) ensures that the power trade between two zones is identical in magnitude and opposite in sign.

3.2.1 Nodal Market Prices

The nodal model is solved for hour $t = 1$. In the base case, the SW is €83,070.22. Transmission line $l25$ (connecting buses 15 and 21) is fully loaded, indicating network congestion. Due to this congestion, LMPs are not identical across all buses, as shown in Figure 5, demonstrating that a single MCP no longer exists when network constraints are included. Nodal prices reflect both generation costs and the impact of transmission constraints. This occurs because supplying an additional unit of demand at a given node may require the redispatch of multiple generators across the network, rather than simply increasing local generation. In the absence of congestion, all nodal prices would be identical and equal to the system marginal cost, as in the copper-plate model.

3.2.2 Sensitivity Analysis

A sensitivity analysis is performed by adjusting the capacity of line $l25$, the most heavily loaded line in the base case. Figure 5 shows nodal prices for three scenarios: base case, reduced (80%), and increased (120%) capacity. Reducing capacity by 20% intensifies congestion and widens the nodal price spread; buses 21 and 22 exhibit lower prices due to access to low-cost generation (g8-g10), but are limited in power export by congestion on $l25$. Conversely, increasing capacity by 20% eliminates congestion, causing all nodal prices to converge to 6.02 €/MWh. The SW follows the same trend: €82,589.26 at 80%, €83,070.22 at 100%, and €83,122.39 at 120%, confirming that transmission capacity directly affects both price formation and market efficiency.

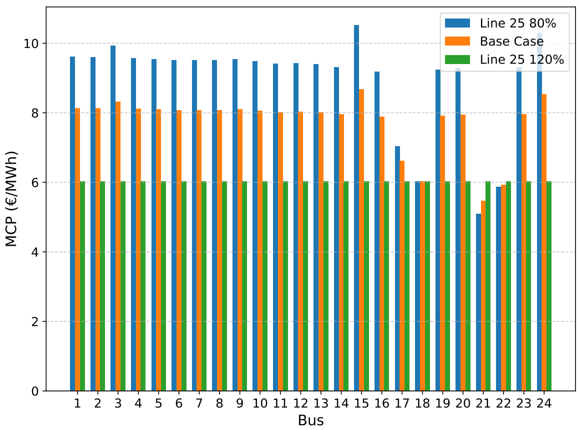

Figure 5: Sensitivity analysis of the nodal market prices at each bus

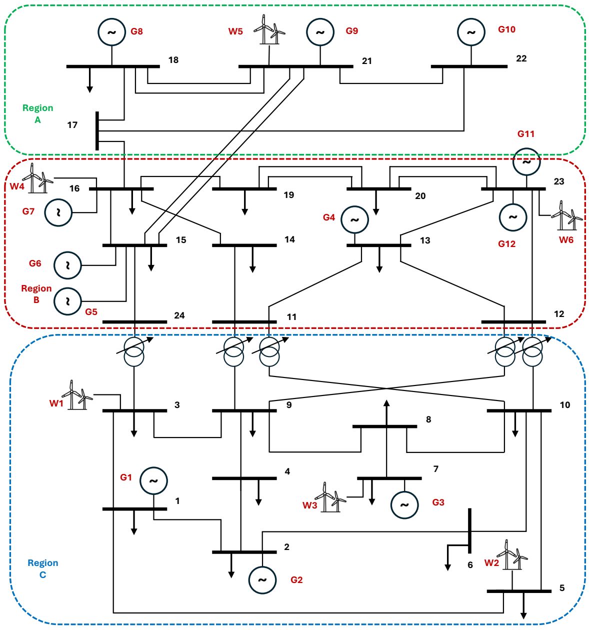

Figure 6: Regions Separation in IEEE 24-bus system

3.2.3 Zonal Market Prices

The zonal pricing framework is implemented by dividing the network into three zones (Figure 6), each assigned a uniform electricity price. The Available Transfer Capacity (ATC) between zones is defined as the total capacity of transmission lines connecting adjacent zones. In this configuration, all three zones yield an identical MCP of 6.02 €/MWh, and the total SW is €83,122.39. The uniform price across all zones suggests that no significant inter-zonal congestion is present under the given network conditions. The power transferred between each zone is $f_{A,B,1} = 688.96\,\text{MW}$ and $f_{B,C,1} = 288.73\,\text{MW}$. This indicates that the power tends to transfer from Region A (Abundant generation) to Region B and Region C, with more demand.

3.2.4 Implications of Nodal vs. Zonal Frameworks

Under nodal pricing, most generators earn zero profit as they are dispatched at marginal cost, with g10 and wind turbines being the main exceptions. Under zonal pricing, the uniform MCP of 6.02 €/MWh slightly increases profits for g9 and g10, but reduces wind revenues, as most wind turbines are located at buses with nodal LMPs exceeding 6.02 €/MWh.

The two models yield notably different outcomes despite operating on the same network. Under nodal pricing, LMPs vary across buses due to congestion on line $l25$, providing granular price signals that reflect the actual physical state of the network. Under zonal pricing, a uniform MCP of 6.02 €/MWh is obtained across all zones, identical to the nodal price observed in the congestion-free sensitivity scenario, suggesting that inter-zonal congestion is not binding, while intra-zonal congestion remains undetected.

From a SW perspective, the zonal model yields a slightly higher value (€83,122.39 vs. €83,070.22 in the nodal model). The difference does not reflect greater efficiency, but rather an overestimation arising from the fact that intra-zonal network constraints are not enforced. Dispatches that appear feasible under zonal pricing may violate internal network constraints, potentially requiring ex-post re-dispatch to restore physical feasibility.

In summary, nodal pricing offers more accurate price signals and a physically consistent dispatch, at the cost of greater market complexity. Zonal pricing simplifies market-clearing but risks masking internal congestion and overestimating SW.

4 Optimization vs. Equilibrium

This section bridges the gap between the centralized social welfare maximization from Step 2 and a decentralized market framework. This part analyzes the behavior of the Energy Storage System (ESS) acting as an independent, price-taking entity to demonstrate that individual profit maximization aligns with the system-optimal outcome.

4.1 Storage Profit-Maximization Problem

In this decentralized setting, the storage operator treats the MCP (λ_t), which corresponds to the dual variable of the power balance constraint in the centralized model, as an exogenous signal. The operator seeks to maximize total profit Π_{ESS} over the 24-hour horizon:

$$\max_{p_t^{ch}, p_t^{dis}, e_t} \quad \Pi_{ESS} = \sum_{t=1}^{24} \lambda_t (p_{dis,t}^{BESS} - p_{ch,t}^{BESS}) \tag{26}$$

Subject to the physical and operational constraints defined in Section 2.1:

$$0 \leq p_{ch,t}^{BESS} \leq \overline{P}_{ch}^{BESS} \qquad (\underline{\mu}_{ch,t}, \overline{\mu}_{ch,t}) \tag{27}$$

$$0 \leq p_{dis,t}^{BESS} \leq \overline{P}_{dis}^{BESS} \qquad (\underline{\mu}_{dis,t}, \overline{\mu}_{dis,t}) \tag{28}$$

$$0 \leq e_t^{BESS} \leq E^{BESS} \qquad (\underline{\mu}_{e,t}, \overline{\mu}_{e,t}) \tag{29}$$

$$e_t^{BESS} = e_{t-1}^{BESS} + p_{ch,t}^{BESS}\eta_{ch} - \frac{p_{dis,t}^{BESS}}{\eta_{dis}} \qquad (\lambda_t) \tag{30}$$

where λ_t is the dual variable for the energy balance constraint, and other parameters remain identical to those used in the centralized problem.

4.2 KKT Conditions and Equivalence Analysis

The Karush-Kuhn-Tucker (KKT) conditions provide the necessary requirements for the decentralized optimal solution.

4.2.1 Stationarity Conditions

Setting the partial derivatives of the Lagrangian function ($\mathcal{L}$) with respect to the decision variables to zero yields the following optimality conditions:

$$\frac{\partial \mathcal{L}}{\partial p_{ch,t}^{BESS}} = -\lambda_t + \eta_{ch}\gamma_t + \overline{\mu}_{ch,t} - \underline{\mu}_{ch,t} = 0 \tag{31}$$

$$\frac{\partial \mathcal{L}}{\partial p_{dis,t}^{BESS}} = \lambda_t - \frac{1}{\eta_{dis}}\gamma_t + \overline{\mu}_{dis,t} - \underline{\mu}_{dis,t} = 0 \tag{32}$$

$$\frac{\partial \mathcal{L}}{\partial e_t^{BESS}} = -\lambda_t + \lambda_{t+1} + \overline{\mu}_{e,t} - \underline{\mu}_{e,t} = 0 \tag{33}$$

4.2.2 Conclusion

The resulting KKT conditions are mathematically identical to those of the social welfare maximization problem from Step 2. This equivalence confirms that a competitive equilibrium, where storage maximizes individual profit based on the system marginal cost, leads to the same dispatch as a central planner. This result validates the efficiency of the market-clearing mechanism in coordinating decentralized flexible assets.

5 Balancing Market

To simulate the balancing scenario, the following assumptions were made:

- Generator g10 is selected as the outage generator, meaning that its day-ahead scheduled production becomes unavailable in real time.
- Wind farms w1–w3 are assumed to produce 10% more than their day-ahead forecast, while wind farms w4-w6 produce 15% less.
- Generators [g1, g2, g3, g4, g7, g11] $\in G_{ser}$ are selected as flexible sources that can provide power regulation.
- Each flexible generator offers upward and downward regulation at the Day-ahead (DA) price plus 10% and minus 15% of its production cost, respectively.

In this step, instead of considering $t = 1$, hour $t = 19$ is selected. This hour corresponds to a peak demand period, while the total wind production is relatively limited. Under these conditions, the outage of generator g10 creates a larger system imbalance that requires additional balancing actions from other generators. For this simulation, the balancing market optimization for one hour is:

$$\min_{p_g^{G\uparrow}, p_g^{G\downarrow}} \quad \sum_g \left((\lambda_{DA} + 0.1\, C_g^G)\, p_g^{G\uparrow} - (\lambda_{DA} - 0.15\, C_g^G)\, p_g^{G\downarrow} \right) + \sum_d 500\, p_d^{D,\text{curt}} \tag{34}$$

$$\text{s.t.} \quad \sum_g \left(p_g^{G\uparrow} - p_g^{G\downarrow} \right) + \sum_d p_d^{D,\text{curt}} = \Delta P \qquad : \lambda_{\text{Balancing}} \tag{35}$$

$$0 \le p_g^{G\uparrow} \le \overline{P}_g^G - P_g^{DA} \qquad \forall g \tag{36}$$

$$0 \le p_g^{G\downarrow} \le P_g^{DA} \qquad \forall g \tag{37}$$

$$0 \le p_d^{D,\text{curt}} \le \overline{P}_d^D \qquad \forall d \tag{38}$$

The objective function minimizes the total cost of balancing actions. Load curtailment is included in the objective function as a penalty cost and also appears in the power balance constraint (Eq. 35). Constraint (36) guarantees that the total generation of each generator (i.e., the sum of the day-ahead scheduled production and the upward regulation) does not exceed its maximum generation capacity. Constraint (37) ensures that the downward regulation provided by a generator cannot exceed its day-ahead scheduled production. This also prevents generators that were not scheduled in the DA market from providing downward regulation. Constraint (38) ensures that the load curtailment of each demand does not exceed its scheduled demand.

5.1 Result

5.1.1 Clearing the balance market

Figure 7 shows the activation of generators in the balancing market for hour $t = 19$. Since the system experiences a power deficit of $\Delta P = 299.1$ MW, only upward regulation is activated. Generator g3 provides 222.43 MW, while generator g1 provides 76.70 MW more, which together cover the imbalance. Therefore, no downward regulation or load curtailment is required. The resulting balancing price is $\lambda_{Balancing} = 15.39$ €/MWh, which is higher than the day-ahead market price $\lambda_{DA,t=19} = 13.32$ €/MWh. This occurs because upward regulation is needed to compensate for the system deficit.

5.1.2 Profit Analysis under Settlement Schemes

Table 5 presents the profits of conventional generators and wind farms under the one-price and two-price imbalance settlement schemes.

For the conventional generators, most units keep the same profit as in the day-ahead market since they do not participate in balancing actions. However, generators g1 and g3, which provide upward regulation, obtain higher profits under the one-price scheme. This occurs because the balancing energy is settled at the balancing price, which is higher than the day-ahead price during a power deficit. Under the two-price scheme, their profits are slightly lower because the

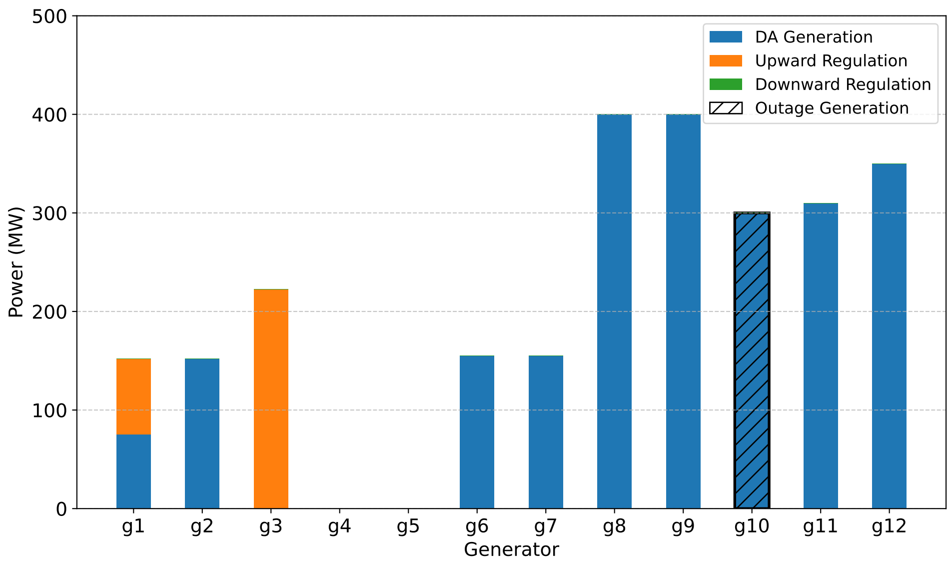

Figure 7: Balancing Market Activation for hour $t = 19$

balancing energy is settled at their regulation bid prices.

Table 5: Profits under different settlement schemes

Generator	**DA [€]**	**1-Price [€]**	**2-Price [€]**
G1	0	1,180.52	1,123.91
G3	0	3,423.19	3,423.19
G6, G7	434.00	434.00	434.00
G8	2,920.00	2,920.00	2,920.00
G9	3,140.00	3,140.00	3,140.00
G10	3,996.00	3,996.00	3,996.00
G11	868.00	868.00	868.00
G12	850.50	850.50	850.50
G2, G4, G5	0	0	0
Total	**12,642.50**	**17,246.21**	**17,189.60**

(a) Conventional Generators

Wind Farm	**DA [€]**	**1-Price [€]**	**2-Price [€]**
W1	639.86	713.78	703.84
W2	1,465.83	1,635.19	1,612.41
W3	763.15	851.33	839.47
W4	649.71	537.11	537.11
W5	593.91	490.98	490.98
W6	592.25	489.61	489.61
Total	**4,704.71**	**4,718.30**	**4,673.42**

(b) Wind Farms

For the wind farms, units W1, W2, and W3 generate positive balancing revenues because they produce more than their scheduled generation, which helps the system during the deficit. This leads to higher profit in the one- and two-price scenario than in the DA market. In contrast, wind farms W4, W5, and W6 incur negative balancing revenues because their production is below the day-ahead schedule. In the two-price scheme, the revenues of wind farms that help the system are lower since their imbalance is settled at the day-ahead price rather than the balancing price. These results show that the one-price scheme provides stronger rewards for balancing services, while the two-price scheme discourages deviations from the day-ahead schedule.

6 Reserve Market

In this step, the model in Step 1 is extended to implement the reserve market in both European-style and US-style frameworks. Two assumptions are applied: the upward and downward reserve requirements are set to 15% and 10% of total demand, respectively, and the generators participating in the reserve market are the same as in the balancing market.

6.1 European scenario

When the reserve and day-ahead markets are cleared sequentially, the reserve market is cleared first. Considering hour $t = 19$, the reserve market problem is as follows:

$$\min_{r^u_{g,t}, r^d_{g,t}} \sum_{g \in G_{ser}} C^u_g r^u_{g,t} + \sum_{g \in G_{ser}} C^d_g r^d_{g,t} \tag{39}$$

Subject to:

$$\sum_{g \in G_{ser}} r^u_{g,t} = 0.15 \sum_d P^D_d \quad : \lambda^u \quad \forall t \tag{40}$$

$$\sum_{g \in G_{ser}} r^d_{g,t} = 0.10 \sum_d P^D_d \quad : \lambda^d \quad \forall t \tag{41}$$

$$0 \leq r^u_{g,t} \leq R^+_g \quad \forall g \in G_{ser}, t \tag{42}$$

$$0 \leq r^d_{g,t} \leq R^-_g \quad \forall g \in G_{ser}, t \tag{43}$$

Objective function (39) minimizes the cost of up and down reserves. Constraints (40) and (41) determine the required reserve power needed. Constraints (42) and (43) limit the power reserved for each generator. The results after optimizing this model are noted as $r^{u*}_{g,t}, r^{d*}_{g,t}$.

After the reserve market is cleared, the DA will be the next market to be optimized. The DA model is similar to the one in Step 1. However, for the generations participating in the reserve market, the power limit constraints have to be modified as follows.

$$r^{d*}_{g,t} \leq p^G_{g,t} \leq \overline{P}^G_g - r^{u*}_{g,t} \quad \forall g \in G_{ser}, t \tag{44}$$

6.2 US scenario (Optional Task)

In contrast to the European (EU) market, the US-style model executes both the reserve and energy markets in the DA stage. Therefore, the problem is now the combination of maximizing the social welfare and minimizing the reserve cost.

$$\max_{P^G_{g,t}, P^W_{w,t}, P^D_{d,t}, r^u_{g,t}, r^d_{g,t}} \left(\sum_{d,t} U_d \cdot p^D_{d,t} - \sum_{g,t} C_g \cdot p^G_{g,t} + \sum_{w,t} 0 \cdot p^W_{w,t}\right) - \left(\sum_{g \in G_{ser}} C^u_g r^u_{g,t} + \sum_{g \in G_{ser}} C^d_g r^d_{g,t}\right) \tag{45}$$

In this case, $r^u_{g,t}$ and $r^d_{g,t}$ are not fixed parameters as in the EU-style DA market, but are optimized with other variables. Other constraints are as in Step 1 and in the European model.

6.3 Results

Table 6 illustrates the difference in reserve prices between the EU and US markets. In the US-style joint clearing, energy dispatch and reserve procurement are optimized simultaneously, so the reserve price reflects not only the reserve bid but also the opportunity cost of withholding capacity from the energy market. This is why US reserve prices are higher than in the EU sequential model, where each market is cleared independently, and this coupling is ignored.

Table 6: Comparison of upward and downward reserve prices between EU and US markets

	Upward reserve price [€]	**Downward reserve price [€]**	**MCP [€]**	**SW [€]**	**Reserve cost [€]**
Only DA	0	0	13.32	109,445.3	0
EU market	17.00	14.00	13.32	107,369.7	6,559.5
US market	19.80	14.61	13.32	108,076.1	7,084.1

The resulting dispatch is shown in Figure 8. Two effects can be observed. First, generators g7 and g11 reduce their energy dispatch due to their upward reserve commitments, since the committed capacity must be held back and cannot be sold as energy. Second, and more importantly, g3 (EU only) and g4 (both EU and US) are forced to operate at a minimum level equal to their downward reserve commitment. This is because downward reserve requires the generator to already be online and generating. As a result, g3 and g4 act as must-run units, crowding out cheaper generators g1 and g2 despite their lower operational costs and available capacity.

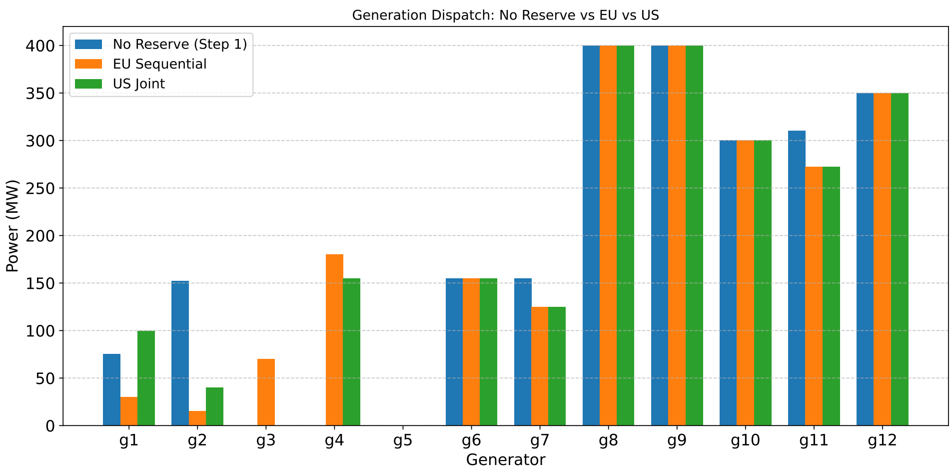

Figure 8: Dispatch power comparison between No reserve market, EU-style reserve market, and US-style reserve market in hour $t = 19$

Despite g3 and g4 operating above their economically justified level, the MCP remains at €13.32 in both models (Table 6). This is because g1 still has remaining dispatchable headroom and continues to be the marginal unit. The presence of must-run generators does not change who sets the price, only how much each generator is dispatched, which is directly reflected in the reduction in SW from the no-reserve case to those in the EU and US markets. This inefficiency is directly reflected in the SW comparison across the three cases. The DA-only model achieves the highest welfare, as no reserve constraints are distorting the dispatch. The US joint model recovers part of the loss by co-optimizing energy and reserve simultaneously, but still falls below the DA-only case since reserve requirements impose a real system cost regardless of how they are allocated. The EU sequential model yields the lowest welfare, as it cannot anticipate how downward reserve commitments force expensive generators into must-run operation, crowding out cheaper units in the subsequent energy market. Despite the lowest welfare, the EU reserve procurement cost is lower than in the US model, since the sequential approach does not account for the opportunity cost of withheld capacity in the reserve price.

References

[1] C. Ordoudis, P. Pinson, J. M. M. González, and M. Zugno, "An Updated Version of the IEEE RTS 24-Bus System for Electricity Market and Power System Operation Studies.", 2016.

[2] M. A. Hannan, S. Wali, P. J. Ker, *et al.*, "Battery energy-storage system: A review of technologies, optimization objectives, constraints, approaches, and outstanding issues", *Journal of Energy Storage*, vol. 42, p. 103 023, 2021.

[3] IRENA, "Electricity Storage and Renewables: Costs and Markets to 2030", International Renewable Energy Agency, Abu Dhabi, Tech. Rep., 2017.

[4] I. Staffell and S. Pfenninger, "Using Bias-Corrected Reanalysis to Simulate Current and Future Wind Power Output", *Energy*, vol. 114, pp. 1224–1239, 2016. DOI: 10.1016/j.energy.2016.08.068.

Appendix A Generation Data

Tables A.1 and A.2 illustrate the generator parameters used in this report. The data is referred to from an IEEE-24 bus system [1]. Note that for each generator, the up and down reserve power are equal (i.e., $R_g^+ = R_g^-$), and similarly the ramp up and down limits are equal (i.e., $R_g^U = R_g^D$). Therefore, only one value is reported for each pair.

Table A.1: Generator parameters: power limits and operation costs

Gen	$\overline{P}_g^G$ (MW)	$\underline{P}_g^G$ (MW)	R_g^+ (MW)	R_g^U (MW/h)	C_g (€/MWh)
g1	152	0	40	120	13.32
g2	152	0	40	120	13.32
g3	350	0	70	350	20.70
g4	591	0	180	240	20.93
g5	60	0	60	60	26.11
g6	155	0	30	155	10.52
g7	155	0	30	155	10.52
g8	400	0	0	280	6.02
g9	400	0	0	280	5.47
g10	300	0	0	300	0.00
g11	310	0	60	180	10.52
g12	350	0	40	240	10.89

Table A.2: Generator parameters: cost coefficients

Gen	C_g^{U} (€)	C_g^{D} (€)	C_g^+ (€/h)	C_g^- (€/h)
g1	15	14	15	11
g2	15	14	15	11
g3	10	9	24	16
g4	8	7	25	17
g5	7	5	28	23
g6	16	14	16	7
g7	16	14	16	7
g8	0	0	0	0
g9	0	0	0	0
g10	0	0	0	0
g11	17	16	14	8
g12	16	14	16	8

Appendix B Wind and Load Data

The wind capacity factor data is obtained from *Renewables.ninja* [4], using five onshore wind locations in Denmark and one in Sweden to match the six wind farms in the model. Since each farm is located at a different geographical position, its capacity factors differ across hours even though all farms have the same installed maximum capacity of 200 MW, reflecting the spatial variability of wind resources. The data corresponds to December 31, 2024, providing hourly capacity factors over a 24-hour period, which are then scaled by each farm's installed capacity to obtain the actual power output. To construct a scenario that better illustrates the role of energy storage in price arbitrage, the hourly sequence is reversed, such that the high-wind period coincides with off-peak demand hours and low-wind production aligns with peak demand. This modification creates a meaningful price spread across hours, allowing the BESS to demonstrate its charging and discharging behavior as intended. Therefore, the power output of each wind farm is calculated and shown in Table B.3 and Figure B.1. The load data is referred to from the IEEE-24 bus system [1], and the total wind power and load profile are illustrated in B.2.

Table B.3: Wind power output (MW)

Hour	w_1	w_2	w_3	w_4	w_5	w_6
1	170.7418	184.2776	186.2220	178.4568	168.2442	179.6832
2	167.6602	182.5956	186.6000	180.0792	170.8332	180.3154
3	162.0560	180.2328	185.7478	179.5802	171.0916	179.3372
4	154.2246	176.5164	184.8418	179.9874	172.7354	178.5290
5	149.0736	171.6846	183.2150	180.2192	174.0610	177.2162
6	144.3468	163.9780	180.6118	179.7864	174.4946	174.6158
7	138.1248	158.0164	176.9856	177.8290	172.3028	171.1846
8	133.1416	155.7284	171.9650	173.3574	163.4586	166.2790
9	132.8870	153.1096	167.3308	167.6956	143.7684	159.0846
10	131.1324	147.3674	161.3940	155.6998	116.4876	147.3892
11	134.6740	142.1418	154.0412	143.2080	88.6904	135.3164
12	130.7198	141.1470	149.4134	124.0094	60.3032	122.5318
13	103.4020	140.2950	145.4288	94.1410	32.9360	106.3158
14	81.0340	143.1074	142.7194	73.8512	39.6926	97.0262
15	53.3024	144.8576	141.5280	52.7396	33.6930	88.8370
16	34.8096	134.5208	141.7600	34.1802	46.7776	78.9324
17	30.3810	117.9450	129.4564	38.4704	41.7934	66.0328
18	38.8150	108.6094	97.9978	30.0012	40.9424	53.5144
19	48.0372	110.0472	57.2938	48.7772	44.5880	44.4634
20	55.1116	113.2934	30.4850	31.5688	34.2826	41.5600
21	64.2296	121.2994	39.4350	33.1970	24.8978	43.4352
22	80.5900	134.4686	20.3252	33.6470	28.3578	47.8534
23	99.4966	151.8408	33.1758	35.6838	26.6300	55.7496
24	119.7036	162.7248	58.3484	24.5812	26.2464	67.5392

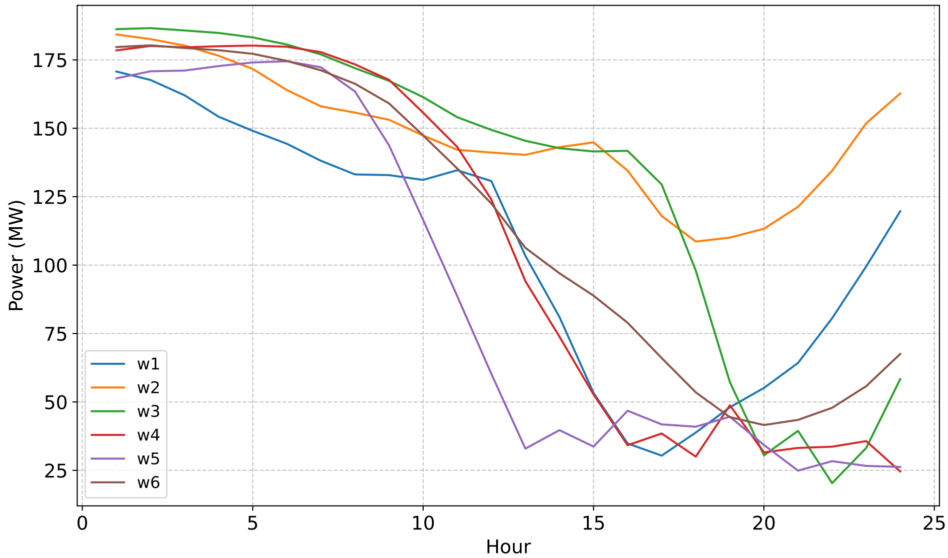

Figure B.1: Hourly wind power of 6 wind farms in 24 hours

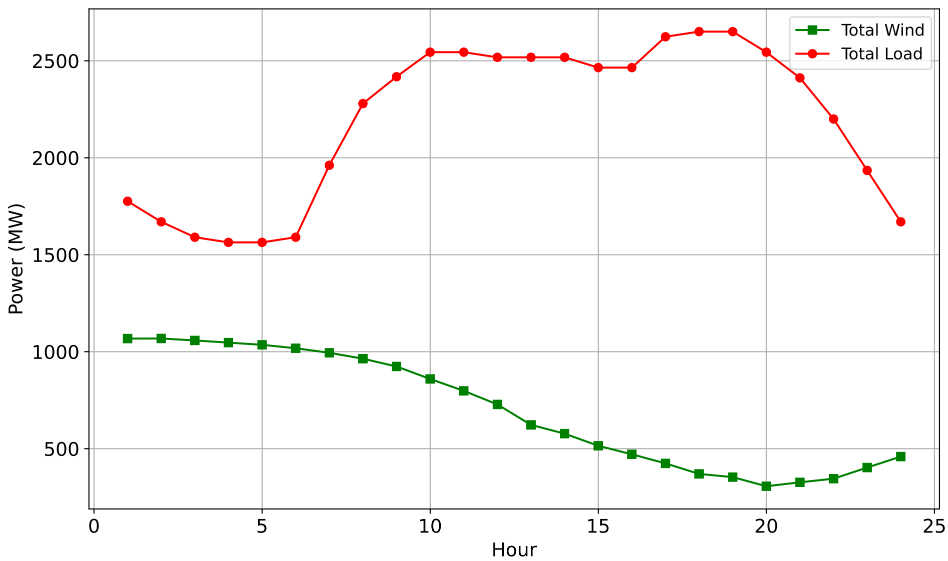

Figure B.2: Wind power and load demand for 24 hours